

VOLUME II

The Database Library

Open data sources, open-source technologies and the communities behind them — what each one gives you, and how to wire it in.

105 sources · 11 shelves

How to use this volume

A catalogue of open data sources, open-source technologies and the communities behind them — satellite imagery, street maps, climate records, conflict data, threat feeds — arranged by shelf rather than alphabetically, because this is a volume you browse rather than look things up in.

Each entry records what the source actually gives you, how to connect to it — the API, the client library or the download route, including whether you need a key — and what it costs. Every entry links to its own project: check each one's licence before you ship.

The companion volume, *The Cyber Dictionary*, defines the vocabulary.

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Satellite & Earth Observation

AWS Open Data Registry *Amazon Web Services*

Hundreds of public datasets hosted free in S3 — Sentinel, Landsat, NOAA weather, OpenStreetMap extracts, genomics, census. [CONNECT](#) Most buckets are anonymously readable: point the AWS CLI or boto3 at them with `-no-sign-request`. Many also publish a STAC catalogue. [ACCESS](#) Free to read [REGISTRY.OPENDATA.AWS](#)

Copernicus Data Space Ecosystem *ESA / European Commission*

Free, full-archive Sentinel-1 (radar), Sentinel-2 (10 m optical), Sentinel-3 and Sentinel-5P imagery for the whole planet, updated every few days. [CONNECT](#) Register for a free account, then use the OData/STAC catalogue APIs to search and the S3-compatible endpoint to download. openEO and Sentinel Hub APIs sit on top for on-the-fly processing. [ACCESS](#) Free [DATASPACE.COPERNICUS.EU](#)

Google Earth Engine *Google*

A planetary-scale analysis platform with a petabyte catalogue already ingested and ready to compute over. [CONNECT](#) Register a cloud project, then use the JavaScript Code Editor or the Python earthengine-api. Free for research, non-profit and education; commercial use is paid. [ACCESS](#) Free for non-commercial [EARTHENGINE.GOOGLE.COM](#)

Maxar Open Data Program *Maxar*

Very high-resolution imagery released free after major disasters and crises, licensed CC BY-NC. [CONNECT](#) Browse the event STAC catalogue and pull cloud-optimised GeoTIFFs directly over HTTP — no key required. [ACCESS](#) Free (non-commercial) [WWW.MAXAR.COM/OPEN-DATA](#)

Microsoft Planetary Computer *Microsoft*

A large STAC catalogue of Earth observation data hosted on Azure, with a hosted JupyterHub for analysis next to the data. [CONNECT](#) Query the public STAC API, then sign asset URLs with the free token endpoint (or the planetary-computer Python package) before reading the COGs. [ACCESS](#) Free [PLANETARYCOMPUTER.MICROSOFT.COM](#)

NASA Earthdata *NASA*

The front door to NASA's entire Earth science archive — MODIS, VIIRS, Landsat, GRACE, SMAP, atmospheric and ocean products. [CONNECT](#) Create a free Earthdata Login, then query the CMR (Common Metadata Repository) API or use the earthaccess Python library, which handles auth and streaming for you. [ACCESS](#) Free [WWW.EARTHDATA.NASA.GOV](#)

NASA FIRMS *NASA*

Near-real-time active fire and thermal anomaly detections worldwide, within about three hours of satellite overpass. [CONNECT](#) Request a free MAP_KEY, then hit the CSV/GeoJSON area API by country or bounding box. WMS layers exist for direct map embedding. [ACCESS](#) Free [FIRMS.MODAPS.EOSDIS.NASA.GOV](#)

OpenAerialMap *Humanitarian OpenStreetMap Team*

Openly licensed drone and aerial imagery, much of it flown after disasters. [CONNECT](#) Search the open catalogue API for imagery footprints, then use the returned TMS tile URL straight in Leaflet or MapLibre. [ACCESS](#) Free [OPENAERIALMAP.ORG](#)

Planet *Planet Labs*

Daily high-resolution commercial imagery. The NICFI programme releases tropical basemaps free for forest and land-use work. [CONNECT](#) Apply for NICFI access, then use the Planet Basemaps API or the XYZ tile endpoint directly in a web map. [ACCESS](#) Free under NICFI, otherwise commercial [WWW.PLANET.COM/NICFI](#)

Sentinel Hub *Planet / Sinergise*

Processing API that renders satellite imagery on demand — pick a bounding box, date range and band maths, get a PNG or GeoTIFF back. [CONNECT](#) OAuth client credentials, then a POST to the Process API with an evalscript. WMS/WMTS endpoints let you drop it straight into Leaflet or QGIS. [ACCESS](#) Free tier, paid above it [WWW.SENTINEL-HUB.COM](#)

UNOSAT *UNITAR*

Satellite-derived damage assessments, flood extents and displacement maps produced for UN operations. [CONNECT](#) Download analysis-ready GIS layers and PDF maps per activation; some layers are served as ArcGIS REST feature services you can consume live. [ACCESS](#) Free [UNOSAT.ORG](#)

USGS EarthExplorer / Landsat *USGS*

Fifty years of Landsat imagery plus aerial photography, elevation and declassified historical satellite data. [CONNECT](#) Free account, then the M2M (machine-to-machine) JSON API for scripted search and bulk download, or pull the same scenes as cloud-optimised GeoTIFFs from AWS Open Data. [ACCESS](#) Free [EARTHEXPLORER.USGS.GOV](#)

Maps & Geospatial

Copernicus DEM / SRTM *ESA / NASA*

Global elevation models — 30 m Copernicus DEM and the older SRTM — for terrain, slope, viewshed and flood modelling. [CONNECT](#) Copernicus DEM tiles are on AWS Open Data as COGs; SRTM comes from NASA Earthdata or OpenTopography's API. [ACCESS](#) Free [SPACEDATA.COPERNICUS.EU/COLLECTIONS/COPERNICUS-DIGITAL-ELEVATION-MODEL](#)

GADM *University of California, Davis*

Administrative boundaries for every country, down to district and ward level. [CONNECT](#) Download per-country GeoPackage or shapefile by URL. Free for academic and non-commercial use — check the licence for anything else. [ACCESS](#) Free (non-commercial) [GADM.ORG](#)

geoBoundaries *William & Mary GeoLab*

Open, CC BY licensed administrative boundaries for every country — the permissively-licensed alternative to GADM. [CONNECT](#) A simple REST API returns a download link per country and admin level; the GeoJSON is small enough to fetch client-side. [ACCESS](#) Free [WWW.GEOBOUNDARIES.ORG](#)

Geofabrik Downloads *Geofabrik*

Daily OpenStreetMap extracts cut by country and region, in PBF and shapefile. [CONNECT](#) Plain HTTP download by URL — trivially scriptable in a build step. Load PBF with osmium or osm2pgsql into PostGIS. [ACCESS](#) Free [DOWNLOAD.GEOFABRIK.DE](#)

GeoNames *GeoNames*

Eleven million place names with coordinates, population, elevation and alternate spellings in many languages. [CONNECT](#) Free account gives you a username to pass to the JSON web services; the full database is also downloadable as tab-separated dumps. [ACCESS](#) Free [WWW.GEONAMES.ORG](#)

Leaflet *Open source*

The small, dependable JavaScript library for raster tile maps and GeoJSON overlays. [CONNECT](#) One script tag or npm install, then `L.map()` and `L.tileLayer()` with any XYZ tile URL. Huge plugin ecosystem for clustering, heatmaps and drawing. [ACCESS](#) Free / open source [LEAFLETJS.COM](#)

MapLibre GL *MapLibre / open source*

The open-source fork of Mapbox GL — fast vector map rendering in the browser and on mobile. [CONNECT](#) npm install `maplibre-gl`, point it at a style JSON and a vector tile source. No API key and no vendor account required. [ACCESS](#) Free / open source [MAPLIBRE.ORG](#)

Natural Earth *Volunteer / NACIS*

Small-scale cartographic base layers — coastlines, borders, rivers, populated places — in three tidy resolutions, public domain. [CONNECT](#) Download shapefiles or GeoJSON directly. Bundled inside most mapping libraries already, so often a one-line import. [ACCESS](#) Free [WWW.NATURALEARTHDATA.COM](#)

Nominatim *OSM Foundation*

Geocoding and reverse geocoding built on OpenStreetMap — address to coordinates and back. [CONNECT](#) The public endpoint allows one request per second with a valid User-Agent. For real volume, run the Docker image against your own extract. [ACCESS](#) Free (rate-limited) [NOMINATIM.ORG](#)

OpenStreetMap *OSM Foundation*

The crowd-built map of the world — roads, buildings, rivers, clinics, shops — open-licensed under ODbL. [CONNECT](#) For live queries use the Overpass API with its own query language; for bulk use Geofabrik regional extracts. Never scrape the main tile server for production. [ACCESS](#) Free WWW.OPENSTREETMAP.ORG

OpenTopography *NSF / UC San Diego*

High-resolution topography, including lidar point clouds, plus hosted global DEMs. [CONNECT](#) Free API key, then the Global DEM REST API returns a GeoTIFF for a bounding box in a single call. [ACCESS](#) Free OPENTOPOGRAPHY.ORG

Overpass Turbo *OSM community*

A browser sandbox for writing and testing Overpass queries against OpenStreetMap before you put them in code. [CONNECT](#) Write the query, run it, export GeoJSON. The generated query string drops straight into a `fetch()` against an Overpass endpoint. [ACCESS](#) Free OVERPASS-TURBO.EU

Overture Maps Foundation *Linux Foundation / Meta, Microsoft, Amazon, TomTom*

Open, standardised global map layers — places, buildings, transportation, administrative boundaries — released quarterly. [CONNECT](#) Data ships as GeoParquet on S3 and Azure. Query it in place with DuckDB's spatial extension, or use the `overturemaps` Python CLI. [ACCESS](#) Free OVERTUREMAPS.ORG

Pelias *Open source*

A self-hostable geocoder that blends OpenStreetMap, OpenAddresses, Who's on First and Geonames into one search index. [CONNECT](#) Run the Docker Compose project for your region, then query its JSON API for search, auto-complete and reverse lookups. [ACCESS](#) Free / open source PELIAS.IO

Protomaps *Protomaps*

A single-file map tile format (PMTiles) that serves an entire basemap from static hosting — no tile server at all. [CONNECT](#) Download or build a `.pmtiles` file, put it on S3 or GitHub Pages, and read it with the `pmtiles` JS library via HTTP range requests. [ACCESS](#) Free / open source PROTOMAPS.COM

Climate & Weather

Climate TRACE *Climate TRACE coalition*

Independently estimated greenhouse gas emissions by asset and sector, built from satellite and sensor data rather than self-reporting. [CONNECT](#) Bulk CSV downloads per country and sector, plus a public REST API for assets and emissions. [ACCESS](#) Free CLIMATETRACE.ORG

Copernicus Climate Data Store *ECMWF / European Commission*

ERA5 reanalysis, seasonal forecasts and climate projections — the standard source for historical hourly weather anywhere on Earth since 1940. [CONNECT](#) Register, accept the licence, then use the `cdsapi` Python client with your key in `~/cdsapirc`. Requests are queued, so plan for asynchronous retrieval. [ACCESS](#) Free CDS.CLIMATE.COPERNICUS.EU

Electricity Maps *Electricity Maps*

Live carbon intensity and power breakdown of electricity grids by zone. [CONNECT](#) Free tier API key covers one zone; commercial plans for more. The historical dataset is also published for download. [ACCESS](#) Free tier, paid above WWW.ELECTRICITYMAPS.COM

Global Forest Watch *World Resources Institute*

Near-real-time deforestation alerts, tree cover loss and land use layers, updated weekly. [CONNECT](#) The GFW Data API serves query and tile endpoints; free API key via the developer portal. Alerts also come as XYZ raster tiles for direct map use. [ACCESS](#) Free WWW.GLOBALFORESTWATCH.ORG

IPCC WGI Interactive Atlas *IPCC*

Regional climate change projections from the assessment reports, explorable by region, scenario and variable. [CONNECT](#) Explore in the browser and export figures and data; underlying CMIP6 data is downloadable from ESGF for analysis. [ACCESS](#) Free [INTERACTIVE-ATLAS.IPCC.CH](#)

NOAA / NWS API *US National Weather Service*

Official US forecasts, alerts, observations and radar as a public JSON API. [CONNECT](#) No key; just set a descriptive User-Agent header. GeoJSON responses drop straight into a map layer. [ACCESS](#) Free [WWW.WEATHER.GOV/DOCUMENTATION/SERVICES-WEB-API](#)

NOAA Climate Data Online *NOAA NCEI*

Station-level historical temperature, precipitation and extremes going back over a century, worldwide. [CONNECT](#) Request a free token by email, then pass it as a header to the CDO v2 REST API. Bulk GHCN files are also on FTP and AWS. [ACCESS](#) Free [WWW.NCEI.NOAA.GOV/CDO-WEB](#)

Open-Meteo *Open-Meteo*

Weather forecast, historical, air quality, marine and climate projection APIs with genuinely simple JSON. [CONNECT](#) No key needed for non-commercial use — a single GET with latitude, longitude and the variables you want. Ideal for prototypes and dashboards. [ACCESS](#) Free (non-commercial) [OPEN-METEO.COM](#)

Our World in Data *Global Change Data Lab / Oxford*

Cleaned, harmonised long-run datasets on climate, energy, health, poverty and more, with the charts to match. [CONNECT](#) Every chart has a download button, and the full data catalogue is on GitHub as tidy CSVs — no key, no auth. [ACCESS](#) Free [OURWORLDINDATA.ORG](#)

World Bank Climate Change Knowledge Portal *World Bank*

Country-level historical climate and projections packaged for planners rather than climate scientists. [CONNECT](#) A documented REST API returns country time series as JSON; the site also exports CSV per indicator. [ACCESS](#) Free [CLIMATEKNOWLEDGEPORTAL.WORLDBANK.ORG](#)

Population & Development

data.gov / data.europa.eu *US Government / European Commission*

National open data portals aggregating hundreds of thousands of government datasets. [CONNECT](#) Both run catalogue APIs (CKAN-style and SPARQL respectively) so you can search programmatically before downloading. [ACCESS](#) Free [DATA.EUROPA.EU](#)

DHS Program *USAID*

Demographic and Health Surveys — nationally representative household data on health, nutrition, gender and mortality. [CONNECT](#) Request dataset access per country with a stated research purpose; the DHS API serves indicator data as JSON without that step. [ACCESS](#) Free (access request) [DHSPROGRAM.COM](#)

Eurostat *European Commission*

Official statistics for the EU and member states — economy, population, environment, transport, at NUTS region level. [CONNECT](#) The Statistics API returns JSON-stat by dataset code; the eurostat Python and R packages make it a one-liner. [ACCESS](#) Free [EC.EUROPA.EU/EUROSTAT](#)

GHSL — Global Human Settlement Layer *European Commission JRC*

Global built-up surface, population grids and settlement classification derived from satellite imagery, back to 1975. [CONNECT](#) Direct download of GeoTIFF tiles by epoch and resolution; also available as WMS layers for quick map use.

[ACCESS](#) Free [HUMAN-SETTLEMENT.EMERGENCY.COPERNICUS.EU](#)

Humanitarian Data Exchange *UN OCHA*

The main open repository for humanitarian data — 20,000+ datasets on displacement, needs, food security, admin boundaries. [CONNECT](#) It runs on CKAN, so the full CKAN API works for search and download. The hdx-python-api package handles pagination and resources. [ACCESS](#) Free [DATA.HUMDATA.ORG](#)

IPUMS *University of Minnesota*

Harmonised census and survey microdata from over 100 countries, made comparable across time and place. [CONNECT](#) Free registration, define an extract in the web interface or via the IPUMS API, then download when it is built. [ACCESS](#) Free (registration) [WWW.IPUMS.ORG](#)

UN Data / SDG Indicators *United Nations*

Official Sustainable Development Goal indicator data by country, target and year. [CONNECT](#) The SDG API returns JSON by series and geo area; bulk downloads are available per goal. [ACCESS](#) Free [UNSTATS.UN.ORG/SDGS/DATAPORTAL](#)

World Bank Open Data *World Bank*

Around 16,000 development indicators for every country — GDP, population, access to electricity, schooling, health. [CONNECT](#) A clean REST API: `/v2/country/{iso}/indicator/{code}?format=json`. No key. The `wbdata` and `wbgapi` Python packages wrap it. [ACCESS](#) Free [DATA.WORLDBANK.ORG](#)

WorldPop *University of Southampton*

Gridded population estimates at 100 m resolution, plus age and sex structure, births and migration. [CONNECT](#) Download GeoTIFFs per country and year, or use the WorldPop REST API to request statistics for your own polygon. [ACCESS](#) Free [WWW.WORLDPop.ORG](#)

Conflict, Rights & Humanitarian

ACLED *Armed Conflict Location & Event Data Project*

Coded records of political violence and protest events worldwide — date, location, actors, fatalities — updated weekly. [CONNECT](#) Register for a free access key, then query the REST API by country, date range and event type. Free for academic and non-profit use. [ACCESS](#) Free for research [ACLEDDATA.COM](#)

Bellingcat Online Investigation Toolkit *Bellingcat*

A curated, maintained index of open-source investigation tools — geolocation, satellite, archives, transport tracking. [CONNECT](#) Browse by category; most entries are free web tools or open-source repos you can run yourself. [ACCESS](#) Free [BELLINGCAT.GITBOOK.IO/TOOLKIT](#)

Forensic Architecture *Goldsmiths, University of London*

Investigations reconstructing human rights incidents from open sources, plus open-source tooling they build to do it. [CONNECT](#) Methods and some tools are published openly on GitHub — notably Timemap for building situated event timelines. [ACCESS](#) Free / open source [FORENSIC-ARCHITECTURE.ORG](#)

IDMC *Internal Displacement Monitoring Centre*

Global estimates of people internally displaced by conflict and disaster, by country and year. [CONNECT](#) The Global Internal Displacement Database offers CSV export and a public API for displacement figures. [ACCESS](#) Free [WWW.INTERNAL-DISPLACEMENT.ORG](#)

IOM Displacement Tracking Matrix *International Organization for Migration*

Field-collected data on displaced populations — locations, numbers, needs — at site and admin level. [CONNECT](#) A public API serves admin-level datasets by country and round; bulk downloads available per operation. [ACCESS](#) Free [DTM.IOM.INT](#)

OpenSanctions *OpenSanctions*

Consolidated sanctions lists, politically exposed persons and watchlists from hundreds of sources, as structured entity data. [CONNECT](#) Bulk JSON and CSV downloads are free under CC BY-NC; a hosted matching API and commercial licence exist for business use. [ACCESS](#) Free (non-commercial) [WWW.OPENSANCTIONS.ORG](#)

ReliefWeb *UN OCHA*

Situation reports, appeals, maps and job postings across every active humanitarian emergency. [CONNECT](#) A well-documented public API supports full-text search and filtering by country, disaster and source. No key; just an appname parameter. [ACCESS](#) Free [RELIEFWEB.INT](#)

UCDP *Uppsala University*

The longest-running armed conflict dataset, with georeferenced events back to 1989 and conflict-year data to 1946. [CONNECT](#) An open REST API returns JSON per dataset version; full CSV and RData downloads need no registration at all. [ACCESS](#) Free [UCDR.UU.SE](#)

UNHCR Refugee Data Finder *UNHCR*

Official refugee, asylum seeker and stateless population statistics by country of origin and asylum, back to 1951. [CONNECT](#) A public REST API serves population and solutions data as JSON; the refugees R package wraps it. [ACCESS](#) Free [WWW.UNHCR.ORG/REFUGEE-STATISTICS](#)

Environment & Biodiversity

Copernicus Marine Service *Mercator Ocean / EU*

Ocean physics and biogeochemistry — sea surface temperature, currents, sea level, chlorophyll — as analysis and forecast products. [CONNECT](#) Free account, then the copernicusmarine Python toolbox to subset and download NetCDF, or open datasets lazily over the ARCO/Zarr endpoints. [ACCESS](#) Free [MARINE.COPERNICUS.EU](#)

GBIF *Global Biodiversity Information Facility*

Over two billion species occurrence records contributed by museums, researchers and citizen scientists. [CONNECT](#) An open REST API for search and a download API for large extracts (free account needed for the latter). pygbif and rgbif wrap both. [ACCESS](#) Free [WWW.GBIF.ORG](#)

Global Fishing Watch *Global Fishing Watch*

Vessel activity and apparent fishing effort derived from AIS tracking, plus port visits and encounters at sea. [CONNECT](#) Register for an API token, then use the vessels, events and 4Wings tile APIs. Map tiles can be layered directly in MapLibre. [ACCESS](#) Free for non-commercial [GLOBALFISHINGWATCH.ORG](#)

Global Flood Monitor / **GloFAS** *European Commission JRC*

Global flood forecasting and monitoring, including river discharge forecasts and rapid flood mapping. [CONNECT](#) Forecast data is distributed through the Climate Data Store API; flood extent layers come as WMS and downloadable rasters. [ACCESS](#) Free [GLOBAL-FLOOD.EMERGENCY.COPERNICUS.EU](#)

IUCN Red List *IUCN*

Extinction risk assessments and range data for over 150,000 species. [CONNECT](#) Request a free API token, then query by species, country or category. Spatial range polygons are a separate download request. [ACCESS](#) Free (token) [WWW.IUCNREDLIST.ORG](#)

OpenAQ *OpenAQ*

Aggregated air quality measurements from government and research monitoring stations worldwide, harmonised into one schema. [CONNECT](#) Free API key, then query measurements by location, parameter and date. The full archive is also mirrored on AWS Open Data. [ACCESS](#) Free [OPENAQ.ORG](#)

Protected Planet / **WDPA** *UNEP-WCMC & IUCN*

The World Database on Protected Areas — boundaries and attributes for every designated protected area on Earth. [CONNECT](#) Free API token for the REST API, or download monthly shapefile and GeoDatabase releases by country. [ACCESS](#) Free [WWW.PROTECTEDPLANET.NET](#)

USGS Earthquake Hazards *USGS*

Real-time global earthquake catalogue, shakemaps and hazard models. [CONNECT](#) The FDSN event API returns GeoJSON with no key; there are also live GeoJSON feeds updated every minute for direct map consumption. [ACCESS](#) Free [EARTHQUAKE.USGS.GOV](#)

Health

DHIS2 *University of Oslo*

The open-source health information platform used as the national system in over 80 countries. [CONNECT](#) Self-host or use the demo server; a comprehensive REST API covers data values, metadata and analytics for integration. [ACCESS](#) Free / open source [DHIS2.ORG](#)

HealthSites.io *Healthsites / OSM community*

An open, validated global registry of health facility locations built on OpenStreetMap. [CONNECT](#) Free API key, then a REST API returns facilities as GeoJSON by country or bounding box. [ACCESS](#) Free [HEALTHSITES.IO](#)

IHME Global Health Data Exchange *Institute for Health Metrics and Evaluation*

Global Burden of Disease results — cause-specific mortality and disability for every country, age and sex, over decades. [CONNECT](#) Use the GBD Results Tool to build a query and export CSV; a results API is available for registered users. [ACCESS](#) Free [GHDX.HEALTHDATA.ORG](#)

OpenFDA *US Food and Drug Administration*

Drug adverse events, recalls, labelling and device reports as queryable open data. [CONNECT](#) Public JSON API with no key for light use; a free key raises the rate limit. Elasticsearch-style query syntax. [ACCESS](#) Free [OPEN.FDA.GOV](#)

WHO Global Health Observatory *World Health Organization*

Official global health statistics — mortality, disease burden, immunisation, health systems — by country and year. [CONNECT](#) The GHO OData API returns JSON by indicator code with no authentication. Athena API serves the same data in XML/CSV. [ACCESS](#) Free [WWW.WHO.INT/DATA/GHO](#)

Economy, Trade & Corporate

GLEIF *Global Legal Entity Identifier Foundation*

The authoritative register of Legal Entity Identifiers, including parent and child ownership relationships. [CONNECT](#) A fully open REST API and daily bulk files — no key, no licence fee, CC0 licensed. [ACCESS](#) Free [WWW.GLEIF.ORG](#)

IMF Data *International Monetary Fund*

Macroeconomic and financial statistics — balance of payments, government finance, exchange rates, WEO projections. [CONNECT](#) An SDMX JSON REST API by dataset and series key, no authentication required. [ACCESS](#) Free [DATA.IMF.ORG](#)

OECD Data Explorer *OECD*

Comparable statistics across member and partner countries — labour, education, tax, environment, innovation. [CONNECT](#) SDMX REST API returning JSON or CSV; the OECD R and Python packages handle the query syntax. [ACCESS](#) Free [DATA-EXPLORER.OECD.ORG](#)

Open Contracting Data Standard *Open Contracting Partnership*

A common schema for public procurement data, used by dozens of national and city contracting portals. [CONNECT](#) Publishers expose OCDS releases as JSON APIs or bulk files; Kingfisher tooling collects and normalises them for analysis. [ACCESS](#) Free [STANDARD.OPEN-CONTRACTING.ORG](#)

Open Ownership *Open Ownership*

Beneficial ownership data — who ultimately controls companies — in a standard data format across jurisdictions. [CONNECT](#) Bulk downloads in the Beneficial Ownership Data Standard JSON; a register API for programmatic lookup. [ACCESS](#) Free [WWW.OPENOWNERSHIP.ORG](#)

OpenCorporates *OpenCorporates*

The largest open database of companies — legal entities, officers and filings from company registers worldwide. [CONNECT](#) REST API with a free key for public-interest and non-commercial use; bulk data under licence for commercial work. [ACCESS](#) Free tier, licensed above [OPENCORPORATES.COM](#)

UN Comtrade *United Nations*

Official bilateral trade statistics — who exports what to whom, by commodity code and year. [CONNECT](#) Free API key from the developer portal, then query by reporter, partner, period and HS code. Rate limits are tight on the free tier. [ACCESS](#) Free tier [COMTRADE.UN.ORG](#)

Security & Threat Data

abuse.ch *abuse.ch / Spamhaus*

Feeds of malware samples, botnet C2 servers, malicious URLs and SSL fingerprints — MalwareBazaar, ThreatFox, URLhaus. [CONNECT](#) Free API key per service, then simple POST queries returning JSON. Plain-text and CSV blocklists are also published for direct ingestion. [ACCESS](#) Free [ABUSE.CH](#)

AlienVault OTX *LevelBlue*

A community threat intelligence exchange — indicators grouped into pulses by contributors. [CONNECT](#) Free account gives an API key; the DirectConnect API pulls subscribed pulses and indicators as JSON. [ACCESS](#) Free [OTX.ALIENVAULT.COM](#)

Censys Search *Censys*

Continuously scanned inventory of internet hosts and certificates — useful for attack surface discovery. [CONNECT](#) Free tier API credentials allow a modest query volume; the REST API and Python client support structured host and certificate search. [ACCESS](#) Free tier, paid above [SEARCH.CENSYS.IO](#)

CIS Benchmarks *Center for Internet Security*

Consensus secure configuration baselines for operating systems, cloud platforms, databases and browsers. [CONNECT](#) Free PDF download after registration; automated assessment content and hardened images are membership or paid. [ACCESS](#) Free (PDFs) [WWW.CISECURITY.ORG/CIS-BENCHMARKS](#)

CISA KEV Catalog *CISA*

Vulnerabilities confirmed to be exploited in the wild — the single best free patch-prioritisation list. [CONNECT](#) One JSON or CSV file at a stable URL, updated as entries are added. No key, trivially automatable. [ACCESS](#) Free [WWW.CISA.GOV/KNOWN-EXPLOITED-VULNERABILITIES-CATALOG](#)

Have I Been Pwned *Troy Hunt*

Breach exposure lookup for email addresses and domains, plus a password-compromise check. [CONNECT](#) The Pwned Passwords range API is free and anonymous via k-anonymity; breach search by email needs a paid key. [ACCESS](#) Mixed [HAVEIBEENPWNED.COM](#)

MITRE ATT&CK *MITRE*

The reference catalogue of real-world adversary tactics and techniques, with groups, software and mitigations mapped to each. [CONNECT](#) The full knowledge base is published as STIX 2.1 JSON on GitHub; the mitreattack-python library loads and queries it locally. [ACCESS](#) Free [ATTACK.MITRE.ORG](#)

NVD *NIST*

The US National Vulnerability Database — CVE records enriched with CVSS scores, affected product identifiers and references. [CONNECT](#) CVE API 2.0 returns JSON; request a free API key to lift the rate limit substantially. Full JSON feeds are downloadable for offline use. [ACCESS](#) Free [NVD.NIST.GOV](#)

OSV *Google / Open Source Security Foundation*

Vulnerability data for open-source packages, keyed precisely to affected versions across every major ecosystem. [CONNECT](#) A free REST API takes a package name and version and returns matching advisories. osv-scanner scans a lockfile or SBOM directly. [ACCESS](#) Free [OSV.DEV](#)

Shodan *Shodan*

A search engine for internet-connected devices and exposed services, with banners, certificates and known vulnerabilities. [CONNECT](#) API key from a paid or academic account; the shodan Python library covers search, host lookup and streaming. [ACCESS](#) Paid, academic access available [WWW.SHODAN.IO](#)

Geospatial Tooling

Cloud-Optimised GeoTIFF *COG community*

A GeoTIFF layout that lets clients read just the tiles they need over HTTP range requests, with no server-side software. [CONNECT](#) Serve a COG from static hosting and read windows of it with rasterio or GDAL — the key to cheap, serverless imagery pipelines. [ACCESS](#) Free / open standard WWW.COGeo.ORG

DuckDB Spatial *DuckDB Foundation*

An in-process analytical database that queries GeoParquet, GeoJSON and shapefiles directly, including files on S3. [CONNECT](#) `INSTALL spatial`; `LOAD spatial`; then query remote Parquet with SQL. The fastest route into Overture data with no infrastructure. [ACCESS](#) Free / open source DUCKDB.ORG/DOCS/EXTENSIONS/SPATIAL

GDAL / OGR *OSGeo*

The translation layer underneath almost every geospatial tool — reads and writes several hundred raster and vector formats. [CONNECT](#) Command line (`gdal_translate`, `ogr2ogr`) or bindings in Python, R and Node. The `/vsicurl/` prefix reads remote files without downloading them. [ACCESS](#) Free / open source GDAL.ORG

Kepler.gl / deck.gl *OpenJS Foundation / Uber*

WebGL layers for rendering very large geospatial datasets in the browser — millions of points, arcs, hexbins and trips. [CONNECT](#) `npm install deck.gl` and compose layers over a MapLibre basemap. Kepler.gl is the no-code application built on top of it. [ACCESS](#) Free / open source DECK.GL

OpenLayers *OSGeo*

A mature JavaScript mapping library with unusually strong support for OGC services, projections and raster analysis. [CONNECT](#) `npm install ol`. Better than Leaflet when you need WMTS, WFS, reprojection or non-Mercator coordinate systems. [ACCESS](#) Free / open source OPENLAYERS.ORG

PostGIS *OSGeo*

Spatial extension for PostgreSQL — geometry types, spatial indexes and hundreds of analysis functions in SQL. [CONNECT](#) `CREATE EXTENSION postgis` on any Postgres instance, then load data with `ogr2ogr` or `shp2pgsql`. [ACCESS](#) Free / open source POSTGIS.NET

QGIS *OSGeo*

The full open-source desktop GIS — analysis, cartography, georeferencing, and a large plugin ecosystem. [CONNECT](#) Install and open. Connects directly to PostGIS, WMS/WFS services and cloud-optimised GeoTIFFs over HTTP; scriptable in Python via PyQGIS. [ACCESS](#) Free / open source QGIS.ORG

Rasterio & Shapely *Open source*

The Python workhorses for reading raster data and manipulating vector geometry respectively. [CONNECT](#) `pip install rasterio shapely geopandas`. GeoPandas ties them together with a familiar DataFrame interface. [ACCESS](#) Free / open source RASTERIO.READTHEDOCS.IO

STAC *STAC community*

SpatioTemporal Asset Catalog — the common JSON standard for describing and searching satellite imagery archives. [CONNECT](#) Learning one STAC API means you can query Copernicus, Planetary Computer, Earth Search and dozens more the same way. Use `pystac-client`. [ACCESS](#) Free / open standard STACSPEC.ORG

Titiler *Development Seed*

A dynamic tile server that turns any cloud-optimised GeoTIFF or STAC item into XYZ map tiles on the fly. [CONNECT](#) Run the Docker image or deploy to Lambda, then request `/cog/tiles/{z}/{x}/{y}` with a `url` parameter pointing at your raster. [ACCESS](#) Free / open source DEVELOPMENTSEED.ORG/TITILER

Communities & Programmes

Datasette *Simon Willison*

A tool for publishing any SQLite database as a browsable, queryable website with a JSON API for free. [CONNECT](#) `pip install datasette`, point it at a `.db` file, deploy anywhere. The fastest way to turn a CSV into a public API. [ACCESS](#) Free / open source DATASETTE.IO

Development Seed Community *Development Seed*

The open-source geospatial community around Development Seed — the team behind STAC tooling, Titiler, eoAPI and much NASA and World Bank open data infrastructure. [CONNECT](#) Follow their open repositories on GitHub, join the community calls and Slack, and reuse their STAC and tiling stacks directly in your own projects. [ACCESS](#) Free

DEVELOPMENTSEED.ORG/COMMUNITY

Humanitarian OpenStreetMap Team *HOT*

Coordinates volunteer mapping of unmapped, vulnerable places, and builds the open tooling to do it. [CONNECT](#) Map through the Tasking Manager, or use their open tools — the Export Tool for custom OSM extracts and OpenAerialMap for imagery. [ACCESS](#) Free WWW.HOTOSM.ORG

Open Data Kit / KoboToolbox *ODK / Kobo Inc.*

Offline-first mobile data collection for field research and humanitarian assessment, used across the aid sector. [CONNECT](#) Free hosted server for humanitarian users, or self-host. A REST API exposes submissions as JSON, CSV or GeoJSON for downstream pipelines. [ACCESS](#) Free tier WWW.KOBOTOOLBOX.ORG

Open Knowledge Foundation *OKFN*

Long-running advocates for open data, and stewards of CKAN, Frictionless Data and the Open Definition. [CONNECT](#) CKAN powers many national data portals, so learning its API unlocks a great many catalogues at once. [ACCESS](#) Free OKFN.ORG

OpenSSF *Open Source Security Foundation*

The cross-industry body working on open-source supply chain security — Scorecard, Sigstore, SLSA, OSV. [CONNECT](#) Run Scorecard against your own repository for an immediate baseline, and use Sigstore to sign releases without managing keys. [ACCESS](#) Free OPENSSE.ORG

OpenStreetMap Foundation *community OSMF*

The forum where OSM tagging, imports and local mapping conventions are actually decided. [CONNECT](#) Read before importing anything at scale — imports without community agreement get reverted. [ACCESS](#) Free COMMUNITY.OPENSTREETMAP.ORG

OSGeo *Open Source Geospatial Foundation*

The umbrella foundation behind QGIS, GDAL, PostGIS, GeoServer and much of the open geospatial stack. [CONNECT](#) Project directories, the OSGeoLive bootable distribution with everything preinstalled, and local chapters worldwide. [ACCESS](#) Free WWW.OSGEO.ORG

Radiant Earth / Source Cooperative *Radiant Earth*

A publishing platform for open geospatial data, hosting large public datasets with permanent, citable access. [CONNECT](#) Browse repositories and read data directly from S3-compatible endpoints; many are STAC-catalogued. [ACCESS](#) Free SOURCE.COOP

Zenodo *CERN*

Open repository for research data and software, issuing a permanent DOI for anything you deposit. [CONNECT](#) A REST API supports programmatic deposit; connect a GitHub repo and every release is archived and given a DOI automatically. [ACCESS](#) Free ZENODO.ORG